

Continuity model runs for the Mediterranean swordfish stock assessment using the a4a initiative

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SCRS/2026/169

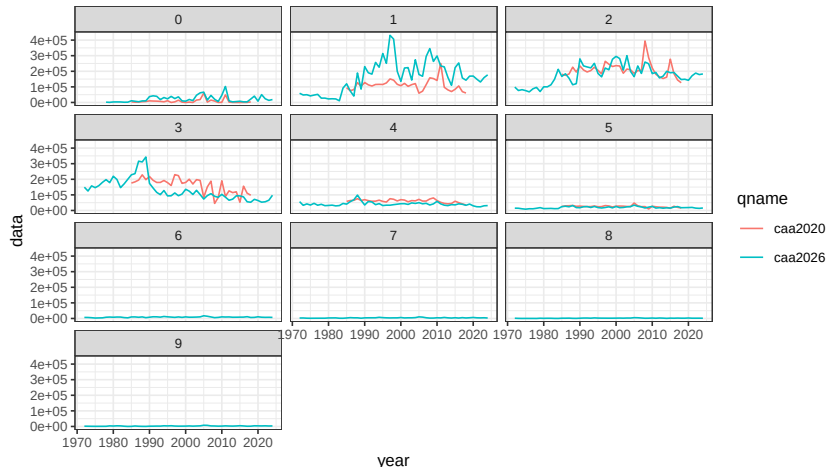
Objectives

To evaluate current stock status by integrating recent fisheries data into the established 2020 4 model configuration.

- Fundamental biological assumptions and model structures remain the same.
- The update focuses on incorporating standardized relative abundance indices and catch-at-age data for 2020–2024, followed by a diagnostic review and revised stock status estimates.

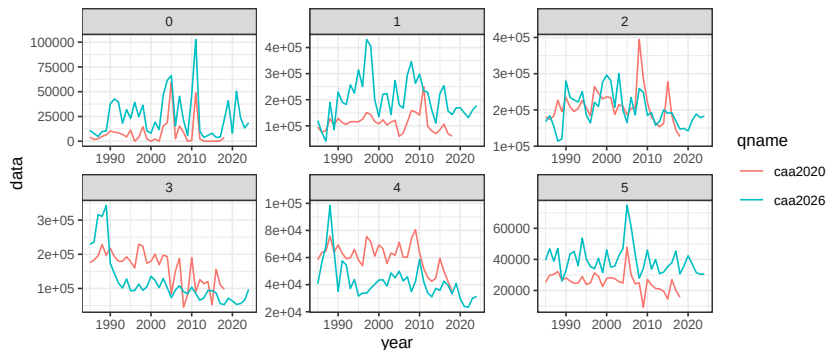
Available catch-at-age data

Catch-at-age (CAA) matrices for ages 0-9 spanning 1972-2024 were provided by the ICCAT Secretariat.



Input catch-at-age data

For continuity with 2020, the fitted model truncates the series to start in 1985 and collapses ages into a plus group at age 5, with the reference fishing mortality ($\bar{F}$) averaged over ages 2-4.



CPUE series

Used in the 2020 assessment:

Index	Series	First year	Last year	Age range
Greek longliners	GR_LL	1987	2024	2-4
Ligurian longliners	LI_LL	2010	2023	2-4
Spanish longliners	SP_LL	1988	2024	2-4
Moroccan longliners	MO_LL	2012	2024	2-4
Sicilian longliners	SI_LL	1991	2009	2-4

Statistical analysis

Four runs are reported:

- The 2020 a4a model as a baseline
- The updated run fitted to data through 2024
- A sensitivity run that truncate the updated data to the 2018, terminal year of the 2020 assessment
- A sensitivity run that truncate the updated data to 2018, using the 2020 standardized indices.

Fishing mortality & recruitment sub-models

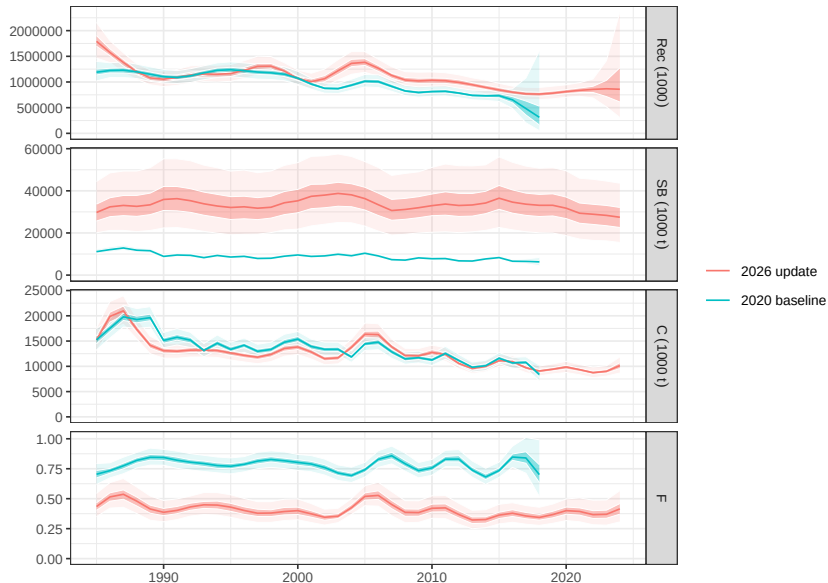
Under the a4a framework, fishing mortality, recruitment and catchability are specified as separable sub-models of smooth functions of age and year.

Run	F submodel	R submodel
2020 baseline	$s(\text{year}, k=17) + s(\text{age}, k=3)$	$s(\text{year}, k=15)$
2026 update	$s(\text{year}, k=19) + s(\text{age}, k=3)$	$s(\text{year}, k=17)$
Sensitivity 1 (up to 2018)	$s(\text{year}, k=17) + s(\text{age}, k=3)$	$s(\text{year}, k=15)$
Sensitivity 2 (up to 2018 - old indices)	$s(\text{year}, k=17) + s(\text{age}, k=3)$	$s(\text{year}, k=15)$

Catchability sub-model

Index	Fleet	qmodel Formula
GR_LL	Greek Longline	~ 1
LI_LL	Ligurian Longline	$\sim s(\text{year}, k = 3)$
SP_LL	Spanish Longline	~ 1
MO_LL	Moroccan Longline	$\sim s(\text{year}, k = 3)$
SI_LL	Sicilian Longline	~ 1

Model results - 2020 vs 2026 update

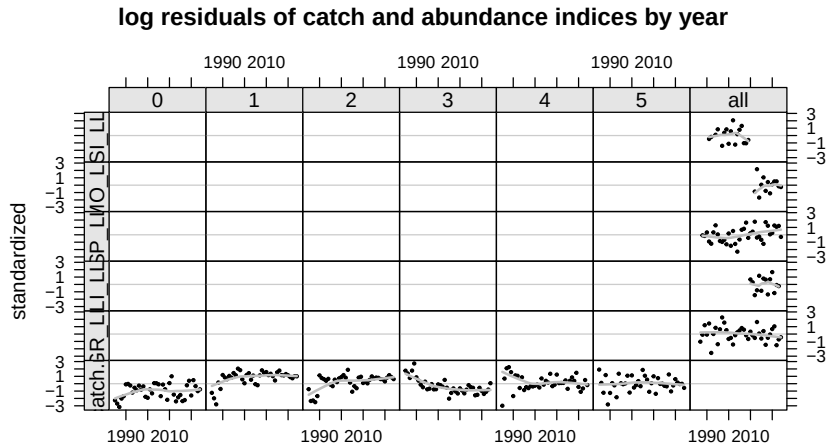


Model results - 2020 vs sensitivity runs



Figure 1: Comparison of the 2020 baseline with the sensitivity runs

Diagnostics (2026 update)



Stock perception: 2020 (left) vs 2026 update (right)

